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A comprehensive review of the impacts of climate change on agriculture in Thailand

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ABSTRACT

The agricultural sector is vulnerable to climate change (CC). Various climate-related extremes, such as droughts, heat waves, unpredictable rainfall patterns, storms, floods, and an increase in insect pests, have adversely affected farmers' livelihoods. Climate forecasts indicate a significant increase in temperatures and more inconsistent, extreme rainfall, obfuscating the prediction of extreme weather events. The IPCC projects that by the end of the 21st century, temperatures in Thailand will rise by 0.95 °C–3.23 °C. This study aims to review the current understanding of CC's impact on the agricultural sector and evaluate the adaptation measures being employed in Thailand. Farmers have begun adopting adaptive measures such as alternative farming techniques, crop diversification, and water management strategies to mitigate climate risks and maintain productivity. However, limited resources, knowledge gaps, and insufficient government support hinder widespread adoption. Targeted interventions and policy support are essential for enhancing adaptive capacity and resilience. The impacts on crop water requirements and livelihoods reveal vulnerabilities due to extreme weather events. Rain-fed agriculture faces significant yield declines and reduced crop water productivity, exacerbating economic impacts on rural households and leading to food insecurity and financial instability. Effective adaptation requires enhanced water management, sustainable practices, and improved institutional support. Community engagement and participatory approaches are vital for building resilience against CC impacts. Comprehensive, region-specific, and long-term studies are crucial for developing robust adaptive strategies.

1. Introduction

The climate is a crucial determinant of agricultural productivity. To identify agriculture's essential role in human survival. Farmers, experts, and stakeholders have expressed concerns about the impacts of climate change (CC), which has promoted vast research on this issue over the past decade (Adams et al., 1998). The IPCC has established that human activities have been the major drivers of observed warming since the mid-20th century. Rising temperatures due to CC are expected to change biophysical interactions in agriculture, affecting growing seasons, species composition, water needs, and soil properties, and increasing risks of pests and diseases (Ullah, 2017). Variations in the rate and intensity of droughts and floods can challenge farmers and threaten food security (Farooq et al., 2022). Water shortages, in particular, can be extremely detrimental (Humphries et al., 2024b). Globally, extreme water scarcity has impacted crops, livestock, and farmers in various regions (Rosa et al., 2020). Scientific evidence suggests that rising temperatures exacerbate these droughts, reduce water supplies, and sometimes intensify wildfires (Wehner et al., 2017). Variations in the rate and intensity of droughts and floods can challenge farmers and threaten food security (Misra, 2014). CC impacts every aspect of the natural environment, primarily transforming the Earth. For example, the temperature rises driven by ecological alterations result in milder winters in some places, enhanced insect pest survivability, and increased plant stress (Skendžić et al., 2021). Crop yields are influenced by temperature fluctuations and the

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List of abbreviations

CC	- Climate Change
IPCC	- Intergovernmental Panel on Climate Change
GHG	- Greenhouse Gas
ONEP	- Office of Natural Resource and Environmental Policy and Planning
NAP	- National Adaptation Plan
CO_2	- Carbon Dioxide
PRECIS RCM	- Providing Regional Climates for Impact Studies – Regional Climate Model
MM5-RCM	- Mesoscale Meteorological – Regional Climate Model
Gt-CO2	- One Billion Tons of Carbon Dioxide
COP3	- Third Conference of the Parties
UNFCCC	- United Nations Framework Convention Climate Change
ET	- Emissions Trading
JI	- Joint Implementation
CDM	- Clean Development Mechanism
NDC	- Nationally Determined Contributions
RE	- Renewable Energy
EE	- Energy Efficiency
CCS	- Carbon Capture Storage
CO2e	- Carbon Dioxide equivalent
ARDL	- Autoregressive Distributed Lag
FDC	- Frequency Domain Causality
RCP 8.5	- Representative Concentration Pathway 8.5
ECHAM4 A2	- ECHAM Climate Model
WII	- Weather Index Insurance
KDML-105	- Khao Dawk Mali 105, one variety of Hom Mali rice
RD-6	- A type of glutinous rice
SEACAM	- Southeast Asia Climate Analysis and Modelling
CanESM2	- Second Generation Canadian Earth System Model
CWR	- Crop Water Requirement
SSP245	- Shared Socioeconomic Pathway 245
IWR	- Irrigation Water Requirement
CWP	- Crop Water Productivity
CWRM	- Community Water-Resource Management
NWRC	- National Water Resource Committee
SCRFD	- Strategic Committee for Reconstruction and Future Development
SCWRM	- Strategic Committee for Water Resources Management
NWRFPC	- National Water Resources and Flood Policy Committee
CWRFM	- Water Resources and Flood Management Committee
ONWF	- Office of National Water Resources and Flood Committee
IWRM	- Integrated Water Resources Management
RBC	- River Basin Committees
TNMC	- Thailand National Mekong Committee
NDPMC	- National Disaster Prevention and Mitigation Committee
EOCs	- Emergency Operation Centers
RID	- Royal Irrigation Department
EGAT	- Electricity Generation Authority of Thailand
NAP	- National Adaptation Plan
NESDB	- Thailand National Economic and Social Development Plans
MNRE	- Ministry of Natural Resource and Environment
CDM	- Clean Development Mechanism
MOAC	- Ministry of Agriculture and Cooperatives

duration of heat or cold waves, with outcomes varying based on plant maturity stages during extreme weather events (Xu et al., 2021). Varied precipitation patterns intensify water scarcity and drought stress for crops, disrupting irrigation water supplies and undermining farmers' planning (Hoffmann, 2013). Additionally, changes in temperature and moisture levels can indirectly influence the absorption rates of minerals and fertilizers, consequently affecting agricultural productivity. So, high temperatures and low precipitation can reduce agricultural productivity below the optimal crop cultivation threshold (Mall et al., 2017). As Asia is home to 4.5 billion people, faces significant climate challenges, especially in rural areas where 75% of the population resides and is most vulnerable to CC impacts (Sterrett, 2011; Habib-ur-Rahman et al., 2022). Despite low greenhouse gas emissions, Asia's economic growth is hindered by CC, exacerbated by the diverse agricultural systems and high susceptibility to flood and drought risks (Lal, 2011; Akram, 2013). Asia is also faced with extreme challenges commencing from CC and variability, as indicated by climatic models forecasting a global mean temperature increase of 1.5 °C between 2030 and 2050, assuming current emission rates persist (Ebi et al., 2018). Projections suggest significant temperature rises in arid regions of western China, Pakistan, and India, alongside heightened intensity of erratic rainfall during monsoon seasons across the region. Additionally, South and Southeast Asia are expected to experience increased aridity due to diminished winter rainfall, while global sea levels are projected to rise by 0.1 m by 2100 (Ebi et al., 2018). Over the past decades, tropical cyclones in the Pacific have surged in frequency and intensity, exacerbating food insecurity in South Asia, home to 262 million malnourished individuals (FAO and IFAD, 2015). Moreover, rural populations in remote drylands and deserts are vulnerable to CC (El-Beltagy and Madkour, 2012). In Asia, CC caused decreasing yields of maize, rice, wheat, potatoes, and vegetables globally, with projections suggesting further serious reductions by 2050 (Mendelsohn, 2014). Therefore, it is imperative to address the threat and impact of CC, which significantly impairs crop production and yield.

In the Asian continent, Southeast Asia is mainly affected by CC (Yang et al., 2021). Southeast Asia, which consists of many island countries, stands out as one of the most vulnerable regions to CC. Beyond geographical factors, the region's populations heavily rely on nature-dependent livelihoods, rendering them susceptible (Lee, 2021). Consequently, these countries face a multitude of climate-induced challenges, including livelihood disruption, food insecurity, disease outbreaks, and forced migration, exacerbating existing poverty and inequality (Jain et al., 2015). Moreover, with over half of the population residing in rural areas engaged in agriculture and related activities (Nykolyuk et al., 2021), the agricultural sector remains pivotal despite its diminishing contribution to GDP and employment. Nevertheless, CC is projected to cause considerable damage to the agricultural economy and farmers in these nations (Singh et al., 2021).

Thailand, renowned as the “Rice Bowl of Asia,” relies heavily on agriculture, a sector employing over 30 percent of the workforce and dominating global exports of rice, rubber, canned pineapple, sugar, fish, tuna, tapioca, and tiger prawns (Lee, 2021). Thailand holds a key position in Southeast Asia's economy and policy-making landscape. As the region's second-largest economy after Indonesia, Thailand wields considerable influence over neighboring countries' development trajectories (Kiguchi et al., 2021). However, environmental risks confronting Thailand's agricultural sector extend beyond domestic concerns, impacting global food security (Attavanich, 2023). Linked by agriculture's dual role as both a victim and contributor to CC, addressing Greenhouse Gas (GHG) emissions from production processes is vital to mitigate its significant contribution to global warming and CC (Kiguchi et al., 2021; Nor Diana et al., 2022). Many initiatives have been implemented to mitigate CC and assist farmers in adapting to changing climatic conditions, with technology adoption emerging as a common adaptation strategy. Lesk et al. (2016) projected a decrease in crop yields, affecting cereal production in the face of extreme heat. The aggravation of the current climate scenarios disrupts farmers' daily lives, altering livelihoods, crop yields, and farming incomes (Lesk et al., 2016). Droughts (Dorman et al., 2013), temperature fluctuations (Lesk et al., 2016), irregular rainfall (Humphries et al., 2024b), and other climatic variables have complex impacts on farmers' lives in Thailand due to CC. The Thai government's Office of Natural Resource and Environmental Policy and Planning (ONEP) created a CC Master Plan in 2015 and a National Adaptation Plan (NAP) in 2019 (Kiguchi et al., 2021). These fluctuations, along with other climatic extremes, directly impact plant and animal production, affecting agricultural productivity through complex environmental interactions (Waibel et al., 2018).

This study aims to critically review and analyze the existing knowledge regarding the effects of climate change on agriculture in Thailand. Current adaptation measures implemented by governmental and agricultural stakeholders. s. The review used a systematic search methodology (Waqas et al., 2023), with keywords specified under relevant topics.

2. Design of the study

This review study examined the recent research findings from different search domains. Systematic reviews proved essential in determining trends and conceptualizing insights from different scientific findings (Waqas et al., 2023). The review process is illustrated in Fig. 1. The review strategy explored the academic view and data support offered by the Science Direct database, Google Scholar, and Scopus databases. Using the selected keywords, the research focused on peer-reviewed publications available through online libraries such as the Science Direct database, IEEE, Google Scholar, and Scopus databases. This review identifies critical knowledge gaps in understanding the localized impacts of specific CC on agriculture in Thailand. There is a lack of empirical data on the effectiveness of adaptation strategies and insufficient research on socio-economic dimensions, particularly for vulnerable populations. Additionally, more integrated research is needed to combine scientific insights with community engagement and policy responses to enhance agricultural resilience.

3. Climate change scenarios and their implications for agriculture

CC scenarios are crucial for understanding how future climate conditions might impact agriculture. These scenarios, based on complex climate models, provide projections of potential climatic changes and their effects on various sectors, including agriculture. The primary tools for projecting future climate conditions are the Representative Concentration Pathways (RCPs) and Shared Socioeconomic Pathways (SSPs) (Humphries et al., 2024a, 2024b). These frameworks help in assessing different GHG emissions trajectories and their potential impacts on climate variables.

The RCPs are a set of GHG concentration trajectories that represent different levels of radiative forcing, a measure of the influence of factors like CO_2 on global temperatures. Introduced by Moss et al. (2010), the RCPs include four pathways: RCP2.6, RCP4.5, RCP6.0, and RCP8.5. RCP2.6 represents a scenario where significant mitigation measures are implemented, resulting in a stabilization of radiative forcing at 2.6 W/m^2 by 2100 (Moss et al., 2010). This pathway anticipates a global temperature increase of approximately 1.0 °C above pre-industrial levels (Rogelj et al., 2016). In contrast, RCP8.5 corresponds to a high-emission scenario with minimal mitigation efforts, leading to radiative forcing reaching 8.5 W/m^2 by 2100 and a global temperature rise of about 3.7 °C (Rogelj et al., 2016). In addition to the RCPs, SSPs provide scenarios for socio-economic developments and their interactions with CC. These pathways help assess how factors such as population growth, economic development, and technological advancements might influence future climate impacts (O'Neill et al., 2014). The integration of SSPs with RCPs allows for a comprehensive evaluation of potential climate impacts under varying socio-economic conditions, providing a nuanced understanding of future climate risks. Agriculture is vulnerable to the changes projected by these scenarios. Temperature increases, for example, can have both direct and indirect effects on crop yields. Higher temperatures can accelerate crop growth but may also lead to heat stress, which reduces yields and affects crop quality. Schlenker and Roberts (2009) found that increased temperatures have a detrimental impact on major crops such as wheat and maize, with the effects being pronounced in tropical regions (Schlenker and Roberts, 2009). Heat stress can impair photosynthesis and increase respiration rates, leading to lower yields and decreased nutritional quality (Lobell et al., 2011). Changes in precipitation can result in increased frequency of droughts or floods, each having profound implications for agriculture. Droughts can reduce water availability for irrigation and decrease soil moisture, leading to lower crop yields. Lobell et al. (2011) highlighted that drought conditions could reduce global maize yields by up to 10% by 2050 (Lobell et al., 2011). Conversely, excessive rainfall can cause soil erosion, nutrient runoff, and waterlogging, negatively impacting crop growth and soil health (Yang et al., 2017) Modeling these impacts involves complex simulations using climate models and agricultural models. Climate models project future climate conditions based on various greenhouse gas emissions scenarios and are crucial for assessing potential impacts on crop yields, water resources, and soil health. The IPCC provides comprehensive assessments of these models and their projections, offering valuable insights for

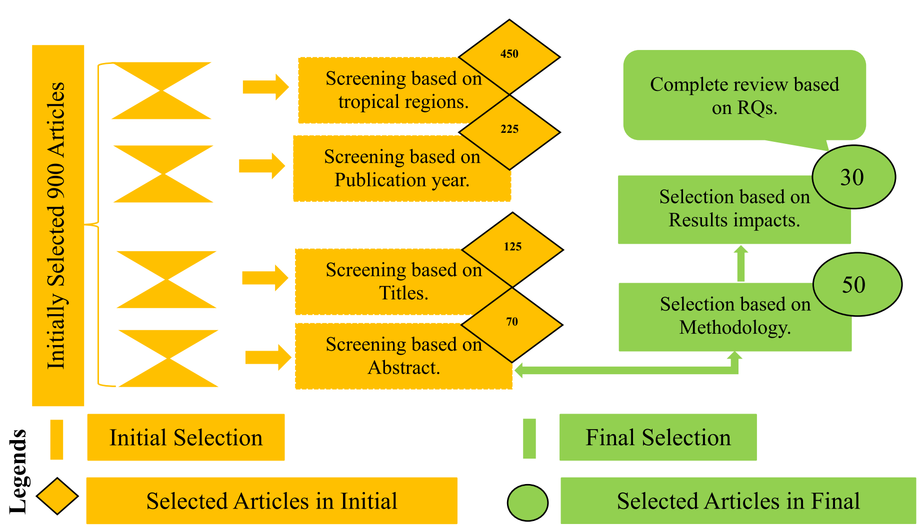

Fig. 1. Overall literature screening process used in the selection of articles.

agricultural planning (Houghton et al., 1990; IPCC, 2007). CC scenarios provide a framework for anticipating and addressing future climate impacts on agriculture. By incorporating these scenarios and associated models, stakeholders can develop informed strategies to mitigate and adapt to CC. Thereby strengthening the resilience of agricultural systems and securing food resources for the future.

4. Geography and hydrometeorology of Thailand

Thailand, situated between latitudes 5°37′ and 20°27′ N and longitudes 97°22′ and 105°37′ E (Fig. 2), has a tropical climate. The country experiences hot temperatures from March to May and a monsoon season from May or June to October, which extends to December in the southern regions (Waqas et al., 2024). Rainfall is crucial for Thailand's agriculture and economy, as 80% of the nation's farming depends on it. Approximately 13,100 acres of farmland in Thailand are located away from irrigated areas (Humphries et al., 2024b). Thailand's geography features diverse climatic regions, each with distinct hydrological characteristics that impact water resources and agricultural practices. Seasonal variations driven by the monsoon cycle significantly influence these regions, causing issues such as flooding, landslides, and droughts at different times of the year (Khedari et al., 2002; Chokngamwong and Chiu, 2006; Sangkhaphan and Shu, 2019). Administratively and geographically, Thailand is divided into five regions: the Northeastern, Northern, Central, Eastern, and Southern regions (Waqas et al., 2024). In the last two decades, Northern Thailand has faced severe impacts from climate change. According to Manton et al. (2001), there has been a significant decrease in the number of rainy days and an increase in the proportion of total rainfall from extreme events in the region (Manton et al., 2001). There has been a significant rise in both minimum and maximum temperatures, an increase in warm nights, and a decrease in cold nights and cool days (Limsakul and Singhruck, 2016; Limsakul, 2020). Tammadid et al. (2023) conducted simulations using the MM5 Regional Climate Model (MM5-RCM), which projects a slight increase in Thailand's average temperature between 2010 and 2029. This warming trend is projected to continue and increase between 2040 and 2059.

Daily precipitation and average temperature data from 1993 to 2022 were obtained from the Thai Meteorological Department and analyzed to identify historical trends and regional impacts. Fig. 2(a) reveals that the Southern Region, the coastal areas, receives the highest precipitation, ranging from 2990 to 5220 mm. The Northern and Northeastern Regions have lower precipitation, between 23.6 and 1370 mm, while the Central Region shows moderate levels, from 1910 to 2470 mm. The Eastern Region also experiences significant precipitation, along the coast. Fig. 2(b) depicts average temperatures, with the Northern Region exhibiting cooler temperatures (16.3–19.8 °C), the Northeastern Region showing warmer temperatures (19.9–21.7 °C), and the Central Region displaying varied temperatures (19.9–24 °C). The Southern Region records the highest temperatures, from 23.1 to 26.6 °C. These climatic trends are essential for understanding the impacts on Thailand's environment and socio-economic activities.

CC, marked by rising temperatures, higher atmospheric CO_2 levels, and altered precipitation patterns, is extensively studied for its impacts. These studies consistently show significant effects on crop production, as

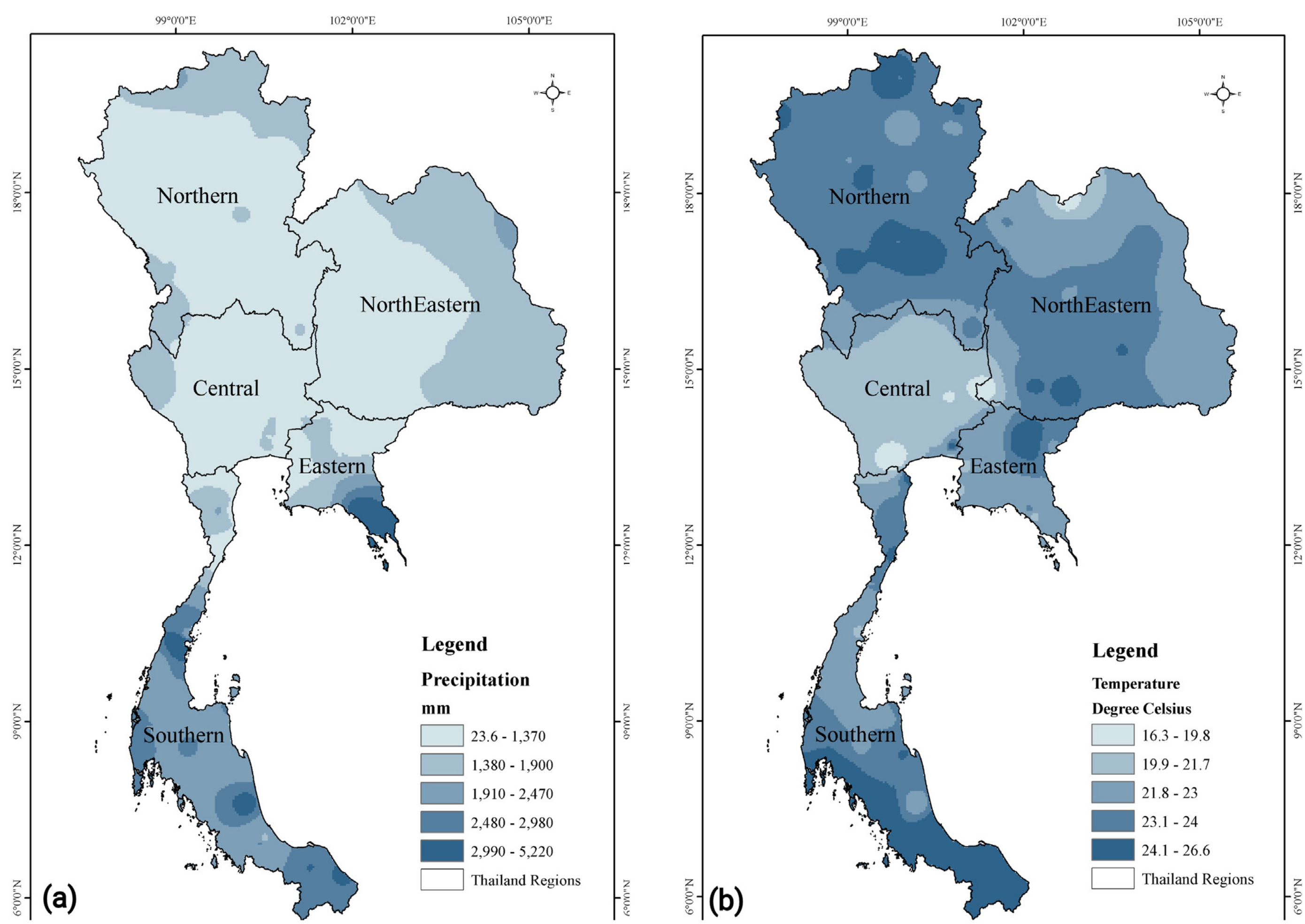

Fig. 2. Climatic regions of Thailand. (a) Average annual precipitation projection for 1993–2022 (b)) Average annual temperature projection for 1993-2022 from Thai Meteorological Department observation station's data.

shown in Fig. 3.

Human-induced GHG emissions, particularly CO_2 from fossil fuel combustion and deforestation have made global warming and CC critical issues in the 21st century (Jain et al., 2015). Continued rises in CO_2 emissions are expected to have profound impacts on the global climate system, influencing all facets of society (Nema et al., 2012). As a result, reducing CO_2 emissions and improving environmental conditions have emerged as global priorities, essential for ensuring sustainable development and alleviating the adverse effects of CC (Nor Diana et al., 2022). Fuel combustion stands out as a primary source of GHG emissions within the energy sector, contributing roughly 32.3 Gt-CO_2, accounting for approximately 96.1% of global CO_2 emissions (Inglesi-Lotz, 2019). In 2016, Asia was the largest GHG emitting region, with 17.4 Gt-CO_2, double that of the US (7.1 Gt-CO_2) and three times that of Europe (5.1 Gt-CO_2). China alone emitted 9.10 Gt-CO_2, accounting for half of Asia's emissions. The Kyoto Protocol, established in 1997, introduced international cooperation mechanisms to aid in emission reduction across two commitment periods: 2008–2012 and 2013–2020 (Misila et al., 2020; Mele et al., 2021). Thailand's energy sector, responsible for 77% of GHG emissions in 2016, is pivotal in emission reduction efforts under the UNFCCC. Embracing renewable energy and enhancing energy efficiency can help Thailand achieve its NDC targets. Attaining 50–75% of the targets outlined in the Renewable Energy (RE) and Energy Efficiency (EE) plans by 2030 is imperative for a 20% reduction in GHG emissions, with a potential 30.4% decrease by 2050 in an extended NDC scenario. Policy recommendations encompass promoting energy efficiency, accelerating renewable energy adoption, and advancing technologies such as Carbon Capture and Storage (CCS) (Misila et al., 2020).

In the agricultural sector, Thailand has 51.3 million hectares of land, with 41% dedicated to various crops; 78% of this agricultural land relies on rainfed farming (Thepent and Chamsing, 2009). Soni et al. (2013) also analyze energy input-output and GHG emissions in Thailand's rainfed integrated agricultural production systems. It finds significant variations in energy use and CO2e emissions across different crops, with cassava consuming the most energy (32.4 GJ/ha) and transplanted rice producing the highest CO_2 emissions (1112 kg of CO_2/ha), due to methane. Major energy inputs are from fossil fuels, particularly diesel (Soni et al., 2013). Research reveals that economic growth, urbanization, industrialization, and tourism each contribute to a 1% increase in CO_2 emissions by 0.97%, 0.17%, 0.06%, and 0.05%, respectively. Conversely, a 1% rise in renewable energy utilization, agricultural productivity, and forest area leads to a decrease in CO_2 emissions by 0.71%, 0.22%, and 0.69%, correspondingly (Raihan et al., 2023). A recent study investigates the interplay between CO_2 emissions, urbanization, energy consumption, and agriculture on Thailand's economic growth spanning 1970 to 2018, employing ARDL and frequency domain causality (FDC) methods. Results suggest a sustained correlation among these factors, with all exerting a positive impact on economic growth. Particularly, the novel application of FDC underscores the predictive capacity of these variables on Thailand's long-term economic performance, emphasizing their substantial implications for macroeconomic indicators (Mata et al., 2021). Hannah (2022) presented data on CO_2 per capita emissions for Asia and Thailand in 2022 (Fig. 4(a)), along with Thailand's share of global CO_2 emissions from 1931 to 2022, as shown in Fig. 4(b).

The review highlights the need for sustainable agricultural practices in Thailand to mitigate greenhouse gas emissions, with studies showing significant variations in energy use and CO_2 emissions across different crops. Targeted interventions for energy efficiency and emissions reduction are essential. These measures will enhance environmental sustainability, economic growth, and resilience.

5. Impacts of climate change on Thailand's agricultural sector

5.1. Impact on agricultural productivity

This section highlights how CC affects agricultural production in different Thai regions. Researchers employ two main methods: statistical modeling and process-based crop modeling to estimate crop productivity under CC impacts. Empirical crop models use statistical relationships to link crop yields to local climate variables (Lobell and Burke, 2010). In contrast, process-based crop models emphasize the physiological processes of crop growth and development in response to climatic factors. These models simulate the entire seasonal crop cycle, encompassing its various stages, to provide a comprehensive understanding (He et al., 2017). In Thailand, limited studies have explored the adverse impacts of CC on rice and other crop production, as summarized in Table 1. For instance, projections suggest a general increase in rice yield under moderate climate scenarios (RCP 2.6, 4.5, and 6.0) across Northeast Thailand throughout the 21st century, with improvements ranging from +0.7% to +22.7% depending on the timeframe. However, under the extreme RCP 8.5 scenario, rice yield initially increases but eventually declines by up to 8.4% in the later part of the century. Northern Thailand

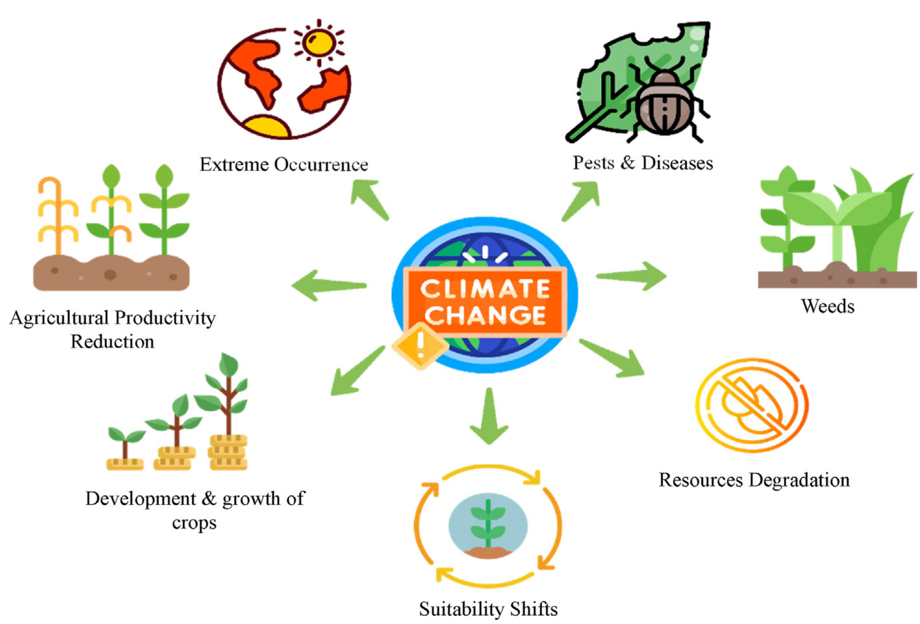

Fig. 3. Impacts of climate change on agriculture.

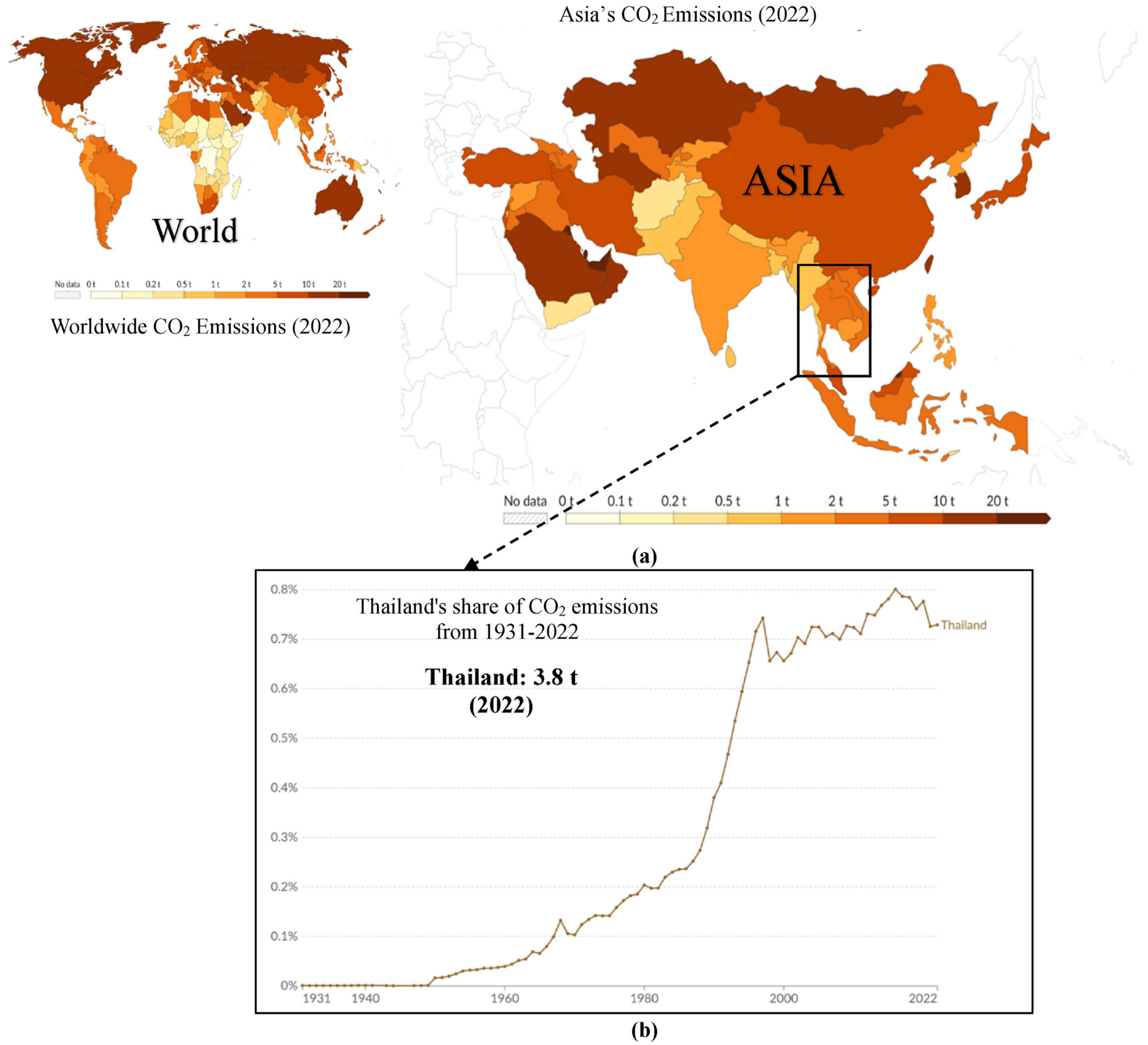

Fig. 4. (a) Per capita CO_2 emissions for World, Asia, and Thailand for 2022 (b) Thailand's share of CO_2 emission from 1931 to 2022 (Source: Hannah, 2022).

faces vulnerability, with rainfed rice production expected to decrease by 5% under RCP 8.5. Similarly, significant declines in rice yield (up to 27.59%) are projected under the ECHAM4 A2 scenario, indicating the adverse effects of increased temperatures on rice productivity. In the Songkhram River Basin, reductions in rainfed rice yields (10–14%) and crop water productivity (CWP) are expected due to increased water stress and temperature changes under RCP 4.5 and RCP 8.5 scenarios. Maize production in Northern Thailand is forecasted to decrease by 4% under the RCP 8.5 scenario, posing a considerable risk to its productivity. For sugarcane, notable yield reductions (23.95%–33.26%) are anticipated, along with decreased harvested area and overall production under both RCP 4.5 and RCP 8.5 scenarios. Under the A2 scenario, oil palm and rubber yields are projected to increase by 2.43%–13.44% throughout the century. Weather Index Insurance (WII) emerges as an effective mitigation strategy to reduce income risk for farmers, particularly enhancing the resilience of rubber and rice production. The impacts of CC on crop yields in Thailand vary, contingent on crop type, regional climatic conditions, and specific climate scenarios. While moderate climate scenarios predict rice yield increases due to enhanced CO_2 concentrations and adaptive practices, extreme scenarios like RCP 8.5 forecast yield declines due to heightened temperatures and altered precipitation patterns. Maize and sugarcane are particularly susceptible to extreme climatic conditions, with significant projected decreases in yield and production area. Conversely, oil palm and rubber demonstrate resilience, with potential yield increases under moderate climate scenarios. Effective adaptation measures, including technological advancements, improved water management, and risk mitigation strategies, are crucial to sustaining agricultural productivity and ensuring food security amidst CC.

5.2. *Impact on water resources and water requirements*

Many studies forecast that CC will affect Thailand's water resources. This section reviews key findings on how CC could impact water

Table 1
Summary of climate change effects on different crop yields in Thailand.

Source	Analysis Period	Study Region	Crop	Impacts on productivity	CC Scenario	Outcomes/Findings
Arunrat et al. (2018)	2020–2039s	Northeast Thailand	Rice	Yield ↑ (+1.3% to +4.5%)	RCP 2.6, 4.5, 6.0, 8.5	Rice yield across all RCP scenarios↑
	2040–2059s			Yield ↑ (+1.1% to +8.6%)		Rice yield across all RCP scenarios↑
	2060–2079s			Yield ↑ (+0.7% to +11.5%), Yield ↓ (−8.4% under RCP 8.5)		Predicted increase in rice yield across most scenarios; ↓ decrease under RCP 8.5
	2080–2099s			Yield ↑ (+2.6% to +22.7%), Yield ↓ (−8.4% under RCP 8.5)		Predicted increase in rice yield across most scenarios; ↓ decrease under RCP 8.5
Arunrat et al. (2021)	2020–2039s	Northern Thailand	Rice	Yield ↑	RCP 2.6, 4.5, and 6.0	Under RCP 2.6, RCP 4.5, and RCP 6.0 scenarios, rice yields ↑
	2040–2059s			Yield ↑	RCP 2.6, 4.5, and 6.0	The rice yields in organic farming are expected to peak, under RCP 4.5.
	2060–2079s			Yield ↑	RCP 2.6, 4.5, and 6.0	Rice yields continue to rise, reaching their highest levels under RCP 4.5.
	2080–2099s			Yield ↓	RCP 8.5	Under RCP 8.5, rice yields are projected to decline significantly, with dramatic decreases in organic carbon and VCSES.
Sinnarong et al. (2022)	2030s	Overall Thailand	Rice, Oil Palm, and Rubber	Rice ↑ 8.14%, Oil Palm ↑ 2.43%, Rubber ↑ 8.89%	A2 Scenario	WII has the potential to mitigate the adverse effects of CC on crop production by reducing income risk for farmers.
	2060s			Rice ↑ 13.37%, Oil Palm ↑ 4.07%, Rubber ↑ 13.15%		The continuation of WII shows increased effectiveness in reducing income risk for farmers, ↑ rubber, and rice.
	2090s			Rice ↑ 13.37%, Oil Palm ↑ 6.48%, Rubber ↑ 13.44%		WII remains effective in mitigating CC impacts, with significant risk reduction observed for all crops.
Amnuaylojaroen et al. (2021)	2020–2029s	Northern Thailand	Rice and Maize	Rice ↓ 5%, Maize ↓ 4%	RCP 8.5	Predicted reduction in rainfed rice and maize production. Maize production is at considerable risk, and rice is at minimal risk.
Pipitpukdee et al. (2020)	2020–2029s, 2046–2055s	Thailand	Sugarcane	yields 23.95–33.26%, harvested area ↓1.29–2.49%, production ↓ 24.94–34.93%	RCP 4.5, and 8.5	Projected decrease in sugarcane yield, harvested area, and production. The largest drop in eastern and lower central regions.
Babel et al. (2011)	2020s,2050s,2080s	Northeast Thailand	Rice	↓17.81%, ↓27.59%, ↓24.34%	ECHAM4 A2 scenario	A decline in rice yield is projected under future climate scenarios.
Boonwichai et al. (2018)	2020s,2050s,2080s	Songkhram River Basin, Thailand	Rice	IWR increase; Rainfed rice yield reduction: 14% (RCP4.5), 10% (RCP8.5)	RCP 4.5, and 8.5	Projected decreases in rainfed rice yield under CC scenarios indicate potential reductions.
Arunrat et al. (2020)	2020–2039s, 2040–2059s, 2060–2079s, 2080–2099s	Overall Thailand	Rice	Rice: ↑ yield under RCP4.5, and ↓ under RCP8.5	RCP 4.5, and 8.5	Under RCP4.5, rice yield ↑ for both farming types under RCP8.5
Sinnarong et al. (2019)	2030s, 2060s,2090s	Thailand	Rice	Mean rice yield decreased by 4.56–33.77%, increased yield variability by 3.87–15.70%.	Future CC projections	A rise in temperature reduces mean rice production and increases production variance
Arunrat and Pumijumnong (2015)	2027s	Thailand	Rice	Under the A2 scenario, rice yield will increase until 2027. The B2 scenario predicts a significant decrease.	A2 scenario, B2 scenario	Under the A2 scenario, rice production and export capacity will expand until 2027, while the B2 scenario predicts a significant production decrease, leading to shortages.

resources and crop demand. Researchers have investigated the effects on various rivers and lake basins, utilizing historical hydro-meteorological data and runoff models alongside global warming scenarios to conduct their analyses Table 2 highlights the CC affects water resources in Thailand, with varying impacts across regions and times. For example, Northeast Thailand faces a substantial increase in the water footprint of rice varieties KDML-105 and RD-6, ranging from 56.5% to 92.2% under RCP 4.5 and 71.4%–76.5% under RCP 8.5 scenarios. Conversely, Chai-Nat-1's footprint may decrease by 42.1%–39.4% under RCP 4.5 and 38.5%–31.7% under RCP 8.5, indicating challenges for agricultural water management due to increased evaporation demand. Average temperatures in the region are expected to rise by 3.1 °C (SEACAM) and 2.2 °C (CanESM2) over the next 30 years, with precipitation projections varying. Groundwater recharge may increase under CanESM2 but slightly decrease under SEACAM while rising groundwater table depths could worsen waterlogging and salinity. In the Songkhram River Basin, temperature increases alongside altered rainfall patterns necessitate adaptation strategies for maintaining agricultural productivity. Land-use changes in the Lower Yom River increase runoff, sediment yield, and nutrient loads, highlighting human impact on hydrological cycles. In the Pak Phanang River Basin, freshwater scarcity and saltwater intrusion diminish rice planting areas, posing food security risks. Effective water management and adaptation strategies are crucial to mitigate these challenges, emphasizing the need for multifaceted approaches in Thailand. The impacts of CC on water resources in Thailand necessitate a multifaceted approach, including improved irrigation systems, sustainable land management, and targeted adaptation strategies to mitigate adverse effects on agriculture and ensure food security.

The effects of CC on CWR across various regions of Thailand are highlighted in Table 3. In the Lower North, predicted increases in rice yields in irrigated areas under the SSP245 scenario contrast with gradual declines in yields projected for mid and far-future periods under SSP585 (Arunrat et al., 2022). It is suggested to include maize, soybean, or mung bean instead of a second rice crop in coffee fields to mitigate negative impacts, particularly in rain-fed areas. Meanwhile, in the Songkhram River Basin, changes in irrigation water requirement (IWR), rice yield,

Table 2
Summary of CC's effect on water resources.

Source	Analysis Period	Study Region	Impacts on productivity	CC Scenario	Outcomes/Findings
Shrestha et al. (2017b)	2020s, 2050s, and 2080s	Northeast Thailand	High increase in water footprint for KDML-105 and RD-6 varieties (56.5%–92.2% under RCP 4.5, 71.4%–76.5% under RCP 8.5), Decrease in water footprint for ChaiNat-1 variety (42.1%–39.4% under RCP 4.5, 38.5%–31.7% under RCP 8.5)	RCP4.5, RCP8.5	High increment in irrigation water requirement due to increased evaporation demand.
Pholkern et al. (2018)	Next 30 years (2045)	Northeast Thailand	Increase in average annual temperature by 3.1 °C (SEACAM) and 2.2 °C (CanESM2) - Decrease in precipitation by 20.85% (SEACAM) and increase by 18.35% (CanESM2)	RCP4.5, RCP8.5	Projected future average annual temperatures are estimated to rise by 3.1 °C (SEACAM) and 2.2 °C (CanESM2), while precipitation is anticipated to decrease by 20.85% (SEACAM) and increase by 18.35% (CanESM2).
Boonwichai et al. (2019)	2030s 2055s 2080s	Songkhram River Basin, Thailand	Changes in maximum and minimum temperatures (+2.8 °C and +3.2 °C under RCP8.5), unchanged annual rainfall, but shifted rainfall patterns	RCP4.5, RCP8.5	Providing irrigation water stands out as the optimal strategy for enhancing rice yield under both RCP4.5 and RCP8.5 scenarios.
Maneepitak et al. (2019)	Current period	Lower Yom River, Thailand	Increased runoff (13–49%), increased sediment yield (37–427%), increased NO3-N load (doubled in upper/middle parts), increased PO4 load (37–377%)	N/A	Land-use change, notably the conversion of forests to agriculture, has resulted in considerable rises in runoff, sediment yield, and nutrient loads.
Pantanahiran (2017)	Current period	Pak Phanang River Basin, Southern Thailand	Impact of environmental changes including freshwater scarcity and saltwater intrusion on rice planting	El Niño, La Niña	The precise effects of El Niño and La Niña on rice production remained unclear. With an annual demand for milled rice at 114,005.63 tons and a production capacity for unhusked rice of 182,516.37 tons/year, there is a minimal surplus (9780.57 tons), leading to low food security. Areas at higher elevations face an increased risk of drought.

Table 3
Summary of climate change's effect on crop water requirement.

Source	Analysis Period	Study Region	Impacts on productivity	CC Scenario	Outcomes/Findings
Arunrat et al. (2022)	(2015–2039), (2040–2069), and (2070–2100)	Lower North Thailand	In irrigated areas, SSP245 forecasts increased rice yields, whereas SSP585 predicts a gradual decline in mid and far-future periods. Additionally, under SSP585, reductions of −6.0% to −17.7% are projected for first and second rice crop yields, contrasting with SSP245's anticipated increase of 3.0%–4.3% shortly.	SSP245, SSP585	To mitigate the adverse effects of CC, it is recommended to consider growing maize, soybean, or mung bean instead of planting a second rice crop.
Humphries et al. (2024b)	2024, 2028, 2033	Northern and Northeastern regions of Thailand	Variations in precipitation patterns, atmospheric water content, and escalating temperatures profoundly influence global agriculture, particularly in tropical regions.	SSP245, SSP370, SSP585	Findings highlight variability in water demand during critical growth stages and the need for farm-specific adaptive strategies.
Boonwichai et al. (2018)	2030s (2020–2044), 2055s (2045–2069) The 2080s (2070–2094),	Songkhram River Basin, Thailand	Changes in IWR, rice yield, and CWP due to CC	RCP4.5 RCP8.5	Under the RCP4.5 scenario, rainfed rice yield may decrease by 14% and CWP could reduce by 32% by the 2080s. Similarly, under the RCP8.5 scenario, rainfed rice yield may decrease by 10% and CWP could decrease by 29% by the same timeframe.
Ngammuangtueng et al. (2023)	2020s and 2030s decades	Central region of Thailand	Lower sugarcane yield, potential feedstock shortage	Increased drought, reliance on groundwater	Sugarcane yields below industrial goal, need for improved water management, energy efficiency, and renewable energy promotion to mitigate climate impacts. Expansion of irrigation is needed for future supply security.
Silalertruksa and Gheewala (2018)	N/A	Chao Phraya and Chi watersheds, Thailand	Implementing irrigation can lead to increased sugarcane yields, with improvements ranging from 23% to 54% compared to non-irrigated systems. Additionally, irrigation can contribute to reduced carbon and ecological footprints of sugarcane products, with reductions of 11%–36% and 15%–35%, respectively.	N/A	Efficient irrigation technology like drip irrigation is crucial for sustainable sugarcane production; Water scarcity potential increases, necessitating management measures for sustainability.

and crop water productivity (CWP) due to CC scenarios RCP4.5 and RCP8.5 indicate potential reductions by the 2080s (Boonwichai et al., 2018; Humphries et al., 2024b). Furthermore, the Central region faces lower sugarcane yields and potential feedstock shortages due to increased drought and reliance on groundwater, necessitating improved water management and energy efficiency measures (Ngammuangtueng et al., 2023). Finally, in the Chao Phraya and Chi watersheds, irrigated sugarcane production systems demonstrate increased yields and reduced carbon and ecological footprints with efficient irrigation technologies, highlighting the need for sustainable water management practices amidst rising water scarcity potential (Silalertruksa and Gheewala, 2018).

5.3. Impacts of climate change on the farmer's livelihoods

Researchers are concerned about the growing threats of CC to the income and consumption patterns of rural communities reliant on natural resources for their livelihood. This section reviews the limited studies available on the economic impacts of CC. For example, Only a few studies on the economic impacts of CC on households in Thailand. Thailand has faced an increase in natural disasters, notably the 2004 tsunami, the 2011 floods, and recent extreme droughts. The 2011 floods, driven by the La Niña phenomenon, inflicted approximately US$ 6.23 billion in damage on the Thai economy, severely impacting the export sector. Major industrial estates in central Thailand were flooded, causing a US$ 2.5 billion drop in exports. This led to significant shortages of consumer goods, pushing headline inflation to 3.9% in 2011. Overall, CC has had a profoundly negative effect on economic growth, worsening livelihoods and damaging the environment (Khambud, 2017; Jatuporn and Takeuchi, 2023). Lertamphainont and Sparrow (2016) analyze the impact of floods and droughts on farming households in Thailand, revealing that crop income significantly declines due to these rainfall shocks. Households cope with excessive rainfall through asset sales and off-farm employment, but droughts are harder to manage, reducing both income and smoothing opportunities. Landless households are particularly vulnerable, struggling more than landholding households to maintain income and consumption levels (Lertamphainont and Sparrow, 2016). A recent study by Jatuporn and Takeuchi (2023) evaluated the impact of CC on the economic growth of Thailand's agricultural sector using data from 1995 to 2019 across 76 provinces. It finds that extreme weather events negatively affect the agricultural economy, while increased rainfall positively influences it (Jatuporn and Takeuchi, 2023). Over the past few decades, despite being a food surplus country, rural households have faced food accessibility issues, exacerbated by rising food prices and production costs. Poor rice farmers, especially those with smaller land holdings, are significantly impacted, facing reduced profits and increased food insecurity (Isvilanonda and Bunyasiri, 2009). Similarly, Bennett (2015) examined how environmental shocks influence rural Thailand using household data. Findings indicate that households with savings or no loans are less likely to use migration as a coping mechanism, while those without savings or with loans are more likely to do so. Policy implications suggest that enabling households to access alternative coping strategies could reduce their reliance on migration during environmental shocks (Bennett, 2015). Also, extreme weather events and deteriorating farming conditions have forced some families to migrate. This migration disrupts community structures and places additional stress on those who remain (Rigg and Salamanca, 2011). Other climate-related also have impacts on farmer's economy, The loss of crop production due to droughts and pest infestations leads to lower quality and quantity of produce, thereby reducing the income of farmers (Norsuwan et al., 2021). The impact of CC on farm families extends beyond financial losses. Health and nutrition are adversely affected as food insecurity becomes more prevalent during prolonged droughts and floods. Families might reduce their food intake, change dietary patterns, or skip meals, leading to malnutrition, especially among children and the elderly (Lin et al., 2022). The increased stress due to CC also leads to social and domestic conflicts. Financial strains, food shortages, and the burden of coping with frequent climate disasters contribute to increased domestic violence and community tensions (Marks, 2011). Moreover, a study explores the impact of various risks on the income of rural households in Pattani province, Thailand, using multiple regression analysis on data from 600 households across 12 districts. Findings indicate that job loss, the death of a household member, and marital problems significantly reduce household income. The study suggests that to mitigate these impacts, the government should create employment opportunities for low-income rural households and establish comprehensive social security programs. Additionally, policies addressing marital issues through holistic family development strategies are recommended (Hardelin and Lankoski, 2015). In Thailand, the impact of CC on the agricultural economy and farmers' livelihoods, as per (Jatuporn and Takeuchi, 2023), is summarized in Table 4.

6. Adaptive measures to climate change

Adaptation is defined as “adaptation to climate variability and change involves ecological, social, and economic systems adjusting to climatic stimuli to mitigate adverse impacts or seize new opportunities” (Hardelin and Lankoski, 2015). The UNFCCC (2007) outlines five essential components: observing climatic and non-climatic variables, assessing impacts, planning, implementing, and monitoring adaptation actions shown in Fig. 5. Additionally, the UNFCCC emphasizes the importance of stakeholder engagement and knowledge management for effective adaptation (Parry et al., 2009). The IPCC has documented that CC is impacting Southeast Asia through rising average temperatures, sea level rise, and alterations in precipitation patterns, though these trends vary significantly across the region (Field et al., 2014). Southeast Asian countries are vulnerable to the negative impacts of global CC, attributable to factors such as their vast coastlines, high concentration of human and economic activities along coastal areas, rapidly expanding populations, and the significant role of agriculture in providing employment and income (Asia, 2009). To reduce the adverse effects of CC, both farmers and government sectors need to adopt adaptive strategies. These measures should not only be technically feasible but also economically viable to ensure their effectiveness (Maneepitak et al., 2019). In the agricultural sector, risks such as droughts and floods can be alleviated through comprehensive risk management at the governmental level and local community water-resource management (CWRM). Sustainable development through CWRM entails: (1) fostering a learning community by leveraging indigenous knowledge and skills, (2) establishing best practice models tailored to various scenarios, and (3) expanding the CWRM network through knowledge and experience sharing (Thanapakpawin et al., 2011).

6.1. Adaptive strategies from the Thai government

6.1.1. Background of water resources and disaster management in Thailand

Thailand's institutional framework for water resources and disaster

Table 4
Impact of climate change on the agricultural economy and farmers' livelihoods in Thailand (Jatuporn and Takeuchi, 2023).

Aspect	Positive Effects	Negative Effects
Rainfall	Positive impact on rainfed crops like rice, cassava, Para rubber, oil palm, and sugarcane.	Rainfall intensification negatively impacts the agricultural economy in both the short and long term.
Temperature	Reduces variability of agricultural growth eventually.	An increase in mean temperatures negatively affects agricultural productivity and revenue in the long term.
Extreme Weather Events	Not impacting positively	Extreme events cause severe damage to agricultural land and production, with substantial economic losses.
Economic Stability	Not impacting positively	CC-induced variability in agricultural productivity causes economic instability for farming households
Income	Not impacting positively	CC reduces farmers' income due to lower productivity and increased crop yield variability.
Livelihood	Not impacting positively	An increase in agricultural households contributes to higher variability of the agricultural economy in the short term.

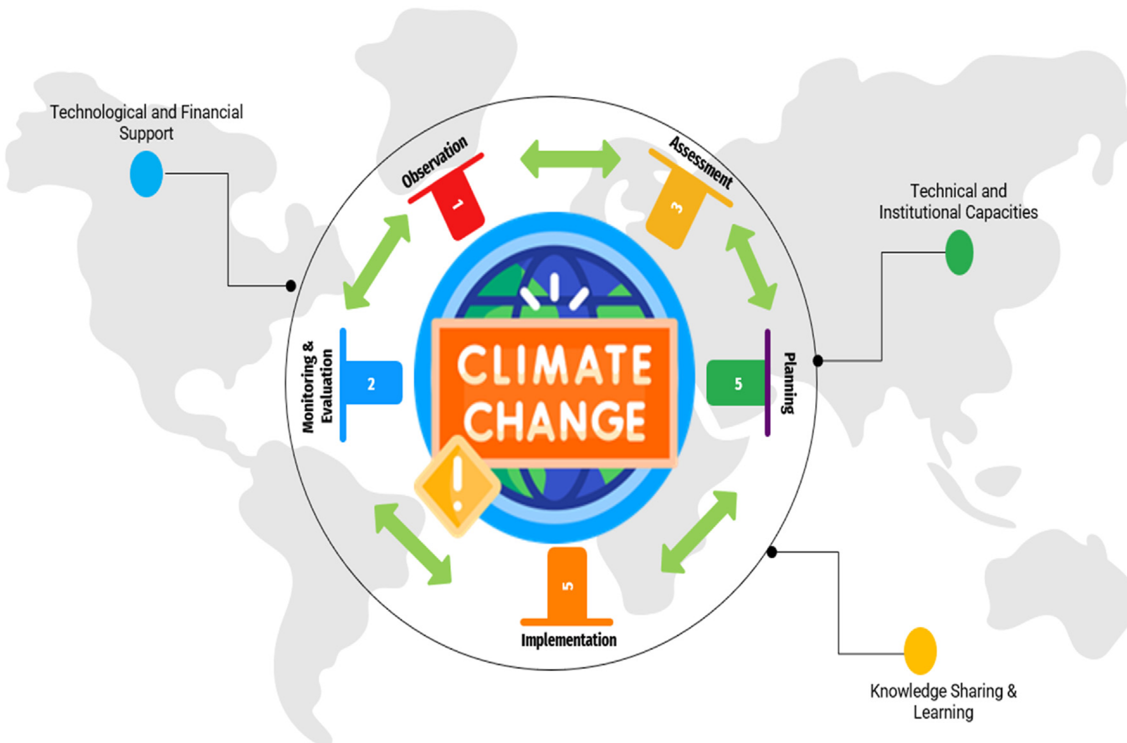

Fig. 5. Five essential components of adaption to CC.

management, shown in Fig. 6, is complex. Responsibility for water resources management is decentralized across multiple committees, ministries, and specialized agencies at national, provincial (76 provinces), and local (district, sub-district, and village) levels. This distribution involves around thirty agencies and bureaus, resulting in considerable fragmentation of roles and responsibilities (Franzetti et al., 2017). The National Water Resources Committee (NWRC), Thailand's initial national water management body, was founded in 1989 and underwent revisions

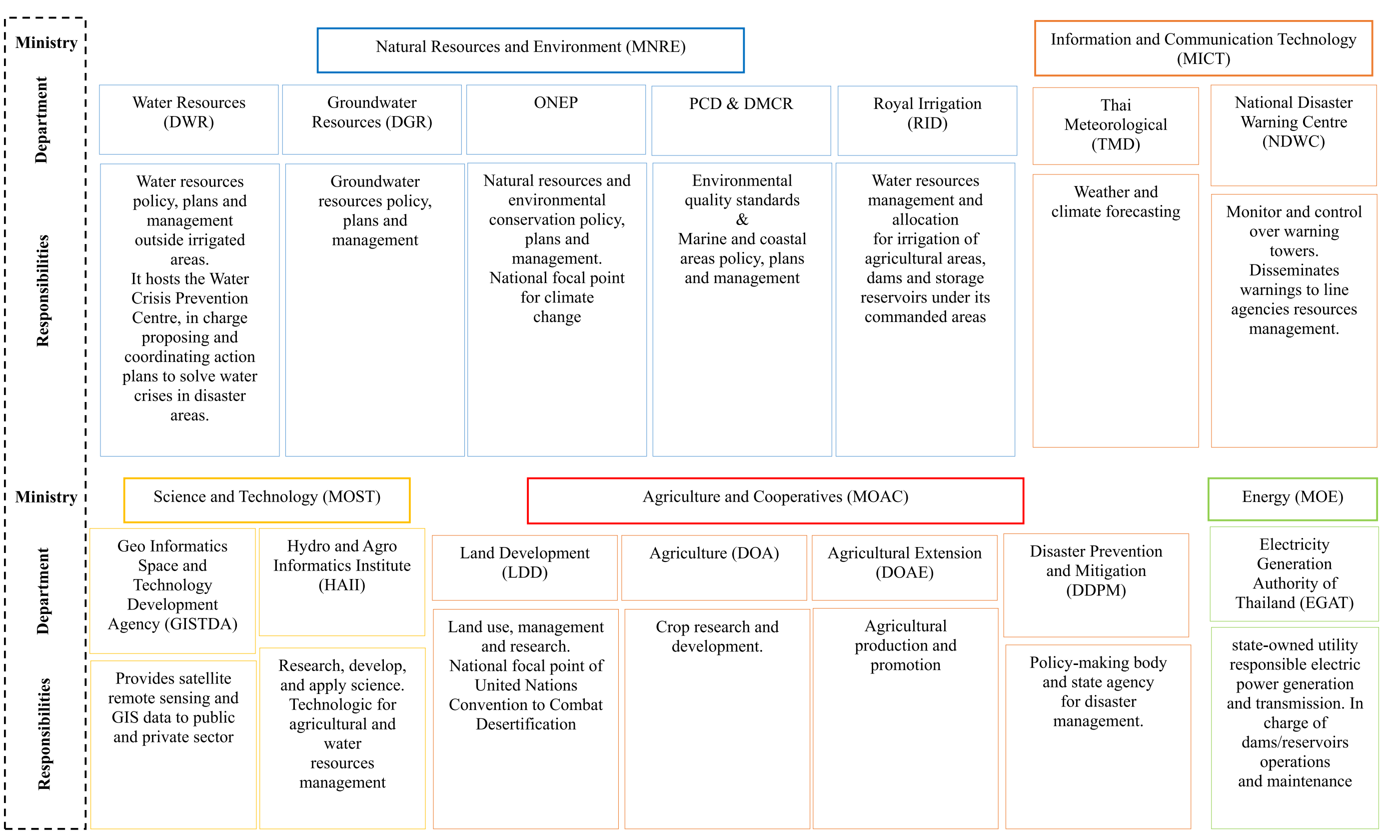

Fig. 6. Thailand's institutional framework governing water resources and disaster management.

in 2002 and 2007. Chaired by the Prime Minister, its duties included overseeing water resources management and policy formulation. However, due to the absence of a legal basis for its permanence, the NWRC underwent frequent changes and lacked substantial authority to enforce policies (Tongdethsri, 2021). Following the significant 2011 flood, Prime Minister Yingluck Shinawatra enacted the Prime Minister's Act on "Reconstruction and Future Development," establishing two committees: the Strategic Committee for Reconstruction and Future Development (SCRFD) and the Strategic Committee for Water Resources Management (SCWRM). The SCWRM was charged with formulating a water and flood management strategy. Additionally, three more bodies were established by another Prime Minister's Act: the National Water Resources and Flood Policy Committee (NWRFPC), the Water Resources and Flood Management Committee (CWRFM), and the Office of National Water Resources and Flood Committee (ONWF), acting as the secretariat and unified command authority (Funatsu et al., 2016) Until May 2014, this committee structure persisted until the Royal Thai Army assumed control of the government, interrupting long-term flood rehabilitation plans initiated by the National Council of Peace and Order. A significant stride towards Integrated Water Resources Management (IWRM) in Thailand occurred with the formation of River Basin Committees (RBCs). The inaugural RBC, the Chao-Phraya River Basin Committee, was established in 1998 (Funatsu et al., 2016). After the establishment of the Department of Water Resources in 2002, twenty-five River Basin Committees (RBCs) were set up. These committees are tasked with crafting water management plans at the basin level, liaising with pertinent agencies, determining water allocation priorities, and overseeing agency performance. RBCs comprise local authorities, water user groups, and community stakeholders (Anukularmphai, 2010a,b). The Thailand National Mekong Committee (TNMC) represents the country within the Mekong River Commission, highlighting transboundary cooperation in the Mekong River Basin. Other pertinent national committees include the National Committee on Climate Change and the National Disaster Prevention and Mitigation Committee, both chaired by the Prime Minister and established in 2006 and 2007, respectively. Disaster management at the provincial level is vital, with provincial governors chairing Disaster Relief Committees and having the authority to declare disaster areas and approve budget requests for local action plans (NDPMC, 2009). During disasters, Emergency Operation Centers (EOCs) are activated at all administrative levels; however, during droughts, their role primarily focuses on monitoring rather than active intervention (interview with DDPM). Civil society stakeholders, including the Thai Red Cross, non-governmental organizations (NGOs), charitable organizations, volunteers, and the private sector, play a pivotal role in supporting local authorities and delivering relief measures during disaster situations (Shaw and Izumi, 2014). In Thailand, water allocation usually takes place annually at the river basin level (Ringler, 2001). In the Chao Phraya river basin, this begins post-monsoon, as central authorities evaluate water levels in major reservoirs and devise pre-seasonal allocation plans (Molle, 2003). Key stakeholders in this process include the Royal Irrigation Department (RID) and the Electricity Generation Authority of Thailand (EGAT), which must collaborate on water release. During crises, inter-ministerial meetings prioritize domestic water consumption (Franzetti et al., 2017; UNESCAP, 2000).

6.1.2. *Overview of adaptative strategies*

The National Adaptation Plan (NAP) of Thailand aims to bolster adaptive capacity and climate resilience across six key sectors: water resources management, agriculture and food security, tourism, public health, natural resources management, and human settlements and security (Asensio et al., 2016). Since 2007, Thailand has prioritized addressing the impacts of CC on agriculture, evident in initiatives like the Thailand National Economic and Social Development (NESDB) Plans 10 and 11, which emphasize managing CO_2 emissions and enhancing biofuel usage. The Ministry of Natural Resource and Environment (MNRE) has actively implemented Clean Development Mechanism (CDM) projects to mitigate CC effects. MNRE outlined six strategies focusing on CC adaptation and mitigation, including vulnerability reduction, carbon sequestration, research and development, public awareness, stakeholder collaboration, and international cooperation frameworks. Additionally, the Ministry of Agriculture and Cooperatives (MOAC) established policies to combat CC impacts, aiming to protect against natural disasters, support affected farmers, ensure food security, and promote livelihood resilience. These measures involve establishing disaster pre-warning systems, providing essential support to farmers, enhancing food security through strategic agricultural production, and facilitating post-disaster recovery efforts (Bantilan et al., 2012). A detailed summary of CC adaption policies focused on agriculture and water is given in Table 5.

In another report by Franzetti et al. (2017), Thailand's response to CC-induced droughts necessitates a shift towards more integrated water resource management. Initial steps include gradual decentralization, establishing main river basin committees with stakeholder representation, and amending laws to empower local authorities. Ongoing political reforms provide an opportunity to advance these reforms, with interim measures such as comprehensive water management plans and expanded support for local water user groups. Ultimately, urgent action is required to ensure sustainable water resources amid increasing demand and climate variability (Franzetti et al., 2017).

Therefore, to improve the adaptive capacity of the agricultural sector in the context of CC, governments must undertake various adaptive actions. Also, farmers must choose appropriate adaptive strategies. If the government and farmers can adapt adequately, Thailand's agricultural sector can withstand changes in temperature and precipitation and take advantage of CC.

6.2. *Adaptive strategies from farmers*

Farmers play a crucial role in addressing CC since adaptation in agriculture primarily occurs at the farm level. To understand human adaptation to CC, it is essential to study farmers' adaptive responses. Farmers can reduce agricultural water demands by altering their farming practices, such as shifting to less water-intensive activities or decreasing cultivated areas (Nasiri et al., 2024). Supplementing agricultural income with non-agricultural pursuits, either on or off the farm, is another tactic (Liu and Zhou, 2023). In some cases, farmers adjust household consumption or resort to borrowing to meet living expenses during drought periods (Applegreen, 2023a,b). Both savings and loans serve as coping mechanisms and investments to mitigate future drought susceptibility (Pienaah and Luginaah, 2024). Farmers' selection of adaptive actions to manage disasters is often influenced by numerous factors, as identified in several studies. These factors typically revolve around farmers' livelihood capitals, including their level of education, farming experience (human capital), sources of income, indebtedness, off-farm earnings (financial capital), farm infrastructure like irrigation systems and electricity (physical capital), land tenure, reliance on forest resources (natural capital), and access to institutions (social capital) (Thanapakpawin et al., 2011; Faisal et al., 2014; Pak-Uthai and Faysse, 2018; Pakuthai, 2019; Yazdanpanah et al., 2023).

However, in Thailand, agriculture is in the initial stages of adaptation. For example, Pak-Uthai and Faysse (2018) examined farmers' short-term adaptive measures in Suphanburi Province, Thailand, during the 2015 and 2016 hydrological droughts. Less than one-third implemented optimal measures with secure water access and market availability, while half resorted to riskier attempts to obtain water or shift production. Other less risky measures, like non-agricultural activities, provided limited income. Overall, most actions did not distinctly increase or decrease vulnerability to drought but varied in the risks they posed (Pak-Uthai and Faysse, 2018). Similarly, Suwanmontri et al. (2018) compared 206 rice-growing households in Northeast Thailand, assessing rainfed rice

Table 5
Summary of climate change adaption policies focused on agriculture and water.

Ministry/Department/ Sector	Policies	Implementations	Year	Strengths	Weaknesses
Ministry of Interior, Department of Disaster Prevention and Mitigation (DDPM)	2P2R measures: Prevention, Preparation, Response, and Recovery	Operated under a single command system, collaboration with the Ministry of Agriculture for artificial rain, water distribution, well digging, canal drainage	2013	Centralized command improves coordination and efficiency	High dependence on central command, which might delay local-specific responses
Ministry of Agriculture and Cooperatives (MOAC)	Crop and livestock management in response to CC	Integrated farming promotion, water resource development in irrigated and non-irrigated zones	Ongoing	Increased resilience in agricultural practices, improved water resource management	Requires continuous monitoring and support for farmers
National Water Resources Management Committee	Enhancing water security and reducing disaster impacts	Development of water security index, early warning systems, integrated risk mapping	2024	A comprehensive approach integrating multiple sectors and advanced technology	Implementation across diverse and remote areas can be challenging
Ministry of Natural Resources and Environment (MNRE)	Supporting mechanisms for water resources management, including ecosystem-based adaptation (EbA) and nature-based solutions (NbS)	Water forecasting systems, river basin management, community engagement in watershed conservation	2024	Focus on sustainable and nature-based solutions enhances long-term resilience	Requires extensive coordination among various stakeholders
Local Administrative Organizations (LAOs)	Participatory policy development for land use and water management	Involvement of local communities in policy-making and conservation efforts	Ongoing	Empowerment of local communities leads to more sustainable and accepted solutions	Varying capacities of LAOs to effectively implement and sustain these initiatives
Academic Institutions	Development of a data warehouse for water management	Research and data collection to inform policy decisions, climate-risk mapping	Ongoing	Evidence-based decision-making improves policy effectiveness	Reliance on continuous funding and data accuracy
Private Sector	Collaboration on water distribution and conservation	Investment in water infrastructure, partnership with academic institutions for data-driven solutions	Ongoing	Additional resources and expertise, innovative solutions driven by private sector efficiency	Profit-driven motives may sometimes conflict with long-term sustainability goals

production using a participatory approach to combat CC. Participants, who were more engaged in local groups and active learning, achieved higher yields and had better knowledge of CC and advanced farming technologies than non-participants. Both groups reacted similarly to past climate damage, but participants had a more cheerful outlook toward adaptation and mitigation. Market-oriented farmers achieved higher yields, indicating that technical improvements in rice production could be enhanced by involving farmers who are community-oriented and eager to learn from researchers (Suwanmontri et al., 2018). Therefore, farmers may need to modify their cropping systems and invest in improved soil and water conservation techniques (Masud et al., 2017). This section reviews a few observed adaptive responses by farmers to climate variability, such as drought (Table 6).

Farmers have implemented adaptive measures across various regions in Thailand to cope with the impacts of CC, including alternative farming techniques, crop diversification, and water management strategies. These efforts have led to improved resilience, sustained agricultural productivity, and enhanced livelihoods despite the challenges posed by climate variability. However, limited resources, knowledge gaps, and inadequate government support hinder the widespread adoption of these measures. Addressing these barriers through targeted interventions and policy support is crucial to further bolstering farmers' adaptive capacity and resilience to future climate risks.

7. Adaptation policies recommendations

This comprehensive review of adaptation measures implemented by government institutions and individuals to mitigate CC impacts on agriculture underscores a clear commitment to resilience. Thailand's adaptation strategies exhibit a multi-faceted approach to addressing drought, characterized by coordinated efforts among various government departments, local authorities, private sector partnerships, and academic institutions. These strategies emphasize enhancing water management, agricultural practices, early warning systems, and community involvement to bolster resilience against drought. Based on this review, the following policies and adaptation options are recommended.

i. Implement integrated farming, crop rotation, and soil conservation techniques to reduce reliance on monocultures and improve soil health.
ii. Stay informed through accurate and timely early warning systems to prepare for and respond to drought conditions effectively.
iii. Develop on-farm small-scale water reservoirs, improve irrigation efficiency, and utilize groundwater management techniques to ensure adequate water supply.
iv. More Engagement with local authorities to align farming practices with agro-economic zoning initiatives to enhance productivity and sustainability.
v. Use climate-resilient crop and livestock varieties developed through research institutions to better withstand changing climatic conditions.
vi. Establish and maintain local food reserves and seed banks to ensure food availability during adverse conditions.
vii. Participate in community-based and cross-sector initiatives to enhance adaptive capacity and share resources and knowledge.

These strategies are designed to provide practical, sustainable solutions that farmers can implement to build resilience against the impacts of CC, particularly droughts.

8. Conclusion and limitations

A comprehensive analysis of the impacts of climate change (CC) on agriculture identifies significant vulnerabilities and adaptation needs. Climate change, characterized by altered precipitation patterns, rising temperatures, and increased frequency of extreme weather events, presents substantial challenges to agricultural productivity and economic stability. While irrigated areas might experience initial productivity gains, rain-fed agriculture—which is vital for Thailand's rural economy—is particularly vulnerable to yield reductions and diminished crop water productivity. The economic consequences for farmers include reduced crop income due to floods and droughts, coupled with rising production costs and food prices, exacerbating food insecurity and financial instability. Coping mechanisms such as asset liquidation, off-

Table 6
Adaptive strategies from farmers to climate variability.

Source	Study Region	Adaptive strategies	Outcomes
Shrestha et al. (2017a)	Doi Mae Salong area, Northern Thailand	• Alternate farming options, and traditional techniques. • Changing crop types and planting times. • Increasing fertilizer use. • Mixed cropping. • Water harvesting structures. • Water rationing, forest protection	• Farmers perceived CC and its impacts; autonomous adaptation is ongoing but limited by lack of knowledge and financial resources; government support is minimal and unplanned
Faisal et al. (2014)	Northeastern Thailand (Ban Khammex, Kalasin Province)	• Adjusting farming practices based on rainfall patterns. • Seeking non-agricultural income. • Community support measures.	• Improved awareness of drought impacts • limited income from non-agricultural activities.
Pakuthai (2019)	Suphanburi Province, Thailand	• Increasing access to water, reducing agricultural water needs. • Obtaining non-agricultural income, by drilling boreholes. • Initiating non-agricultural activities	• One-third implemented low-risk measures to maintain the usual income. • Half chose high-risk measures with potentially high returns. • Others opted for less risky measures with limited income; limited public support impact
Polthanee et al. (2014)	Nakhonratsima and Kalasin, Northeast Thailand	• Crop diversification. • Change of land use pattern • Building on-farm ponds	• Implemented adaption measures to overcome the yield loss in previous years when rainfall reduced in the drought year 2012 compared to the 2001–2011 average.
Babel et al. (2024)	Mun River basin	• Adoption of local adaptation measures by farmers to cope with droughts; diversifying crops, building farm ponds, and implementing crop management practices.	• Farmers in Dan Khun Thot district have implemented diversified crops and various adaptive measures, resulting in low vulnerability and risk. • Conversely, adaptation measures in the Phlapphla Chai district are less prevalent, leading to higher vulnerability and risk levels.

farm employment, and migration underscore the difficulties in maintaining livelihoods. Vulnerable populations, including landless and resource-poor households, face intensified inequities. Governmental adaptive measures focus on enhancing water management, promoting sustainable agricultural practices, and establishment of early warning systems. However, fragmented institutional frameworks pose challenges to effective implementation. Farm-level adaptations involve practice adjustments, crop diversification, and improved water use efficiency. This review highlights the following key findings:

- Significant impacts of variations in precipitation, rising temperatures, and extreme weather on agricultural productivity and economic stability.
- Severe economic effects on farmers' income, increased production costs, and food prices contribute to food insecurity and financial instability.
- Asset sales, off-farm employment, and migration as responses to climatic unpredictability.
- Exacerbation of inequities among landless and resource-limited households.
- Government initiatives in water management and sustainable practices, and farm-level adaptations including practice adjustments and improved water use efficiency.

The study relies on existing literature and secondary data, which may not fully capture recent developments or localized impacts of climate change. Additionally, the analysis is constrained by variability and uncertainty in climate projections and does not address socio-cultural dimensions critical for effective adaptation. Evaluation of government and farmer initiatives is also limited by the availability of comprehensive, long-term empirical data.

Data availability statement

Data used to support the study's findings can be obtained from the corresponding author upon request.

CRediT authorship contribution statement

Muhammad Waqas: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Adila Naseem:** Writing – review & editing, Investigation, Data curation. **Usa Wannasingha Humphries:** Writing – review & editing, Supervision, Resources, Project administration. **Phyo Thandar Hlaing:** Writing – review & editing, Visualization, Validation. **Muhammad Shoaib:** Writing – review & editing, Supervision, Conceptualization. **Sarfraz Hashim:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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